

Identifiability and Feasibility Budgets in Pre-Export Row Scaling for Mixed-Integer Programs

Mathias Rodríguez Castro

Facultad de Ingeniería, Universidad de la República, Montevideo, Uruguay

mathiasr@fing.edu.uy

Abstract

Positive row scaling preserves a mixed-integer linear program in exact arithmetic, but an observed numerical improvement need not identify the information used to select the factors. We analyze a local-block-coupling construction and prove that geometric-mean row centering can erase its block-scale information exactly. With a near-identity band, the remaining scale information is bounded independently of the original heterogeneity. A matched kernel control explains an apparent order-of-magnitude spectral improvement without using block scales. We then characterize the positive row factors compatible with an explicit original-unit residual budget and a coefficient window. The admissible set is an interval; constrained geometric-mean centering is its logarithmic minimax solution, and an empty interval provides an explicit incompatibility certificate. An integer-rounding bound isolates an error that scaling cannot suppress, and a discrete characterization selects exactly representable power-of-two exports. A reconstruction of 240 dispatch runs separates completed-run effort from failures, and 3,456 configurations of exactly enumerable binary stress instances distinguish conditional row guarantees from solver behavior. Budget projection improves verification on HiGHS in this stress experiment, but does not establish a solver-independent advantage; a rounding-aware contract trades acceptance for conservative abstention. The results provide precise controls for attributing scaling gains and for stating what a numerical scaling policy can actually guarantee.

Keywords: mixed-integer linear programming; row scaling; numerical reliability; equilibration; residual budgets; computational reproducibility.

1 Introduction

Model generators retain variable ownership, constraint roles and application units that a conventional matrix export may discard. Such information can guide scaling, but its presence in an algorithm does not establish that it affects the exported matrix. Row coverage, normalization kernels, safeguards and metadata can change together, confounding the interpretation of a numerical improvement.

In mixed-integer linear programming (MILP), positive row scaling preserves the exact feasible set while changing the representation seen by presolve, relaxation solves and integer acceptance tests. Coefficient magnitudes, matrix condition numbers, original-data residuals and running time are therefore distinct outcomes. We study which generator information remains identifiable after preprocessing and which numerical guarantees can be stated independently of solver behavior.

First, we analyze a local-block-coupling construction whose block statistic is the geometric mean of the smallest and largest post-local coefficient magnitudes. Exact geometric-mean row centering makes every such statistic one; an identity band permits only bounded variation. A matched Euclidean coupling-row control then explains an apparent block-metadata benefit. Counterexamples and direct export comparisons delimit the result when practical safeguards intervene.

Second, an original-unit residual budget supplies information that survives a change of row units when transformed with the row. We characterize the factors that combine this budget with coefficient limits, derive the constrained logarithmic minimax factor, and extend admissibility to integer rounding and exact power-of-two export. An empty admissible set certifies incompatible numerical specifications, not MILP infeasibility. Solver acceptance remains a separate question.

The evidence combines a controlled matrix audit, reconstructed dispatch measurements, and exactly enumerable binary stress instances. Direct tolerance scaling and matched kernel controls prevent crediting the proposed construction with benefits already explained by simpler rules. The contribution is a characterization and reproducible audit of these mechanisms, with explicit limits on the guarantees.

2 Related work

Diagonal scaling has a long history, including the logarithmic method of Curtis and Reid [3]. Elble and Sahinidis [5] compare scalers for simplex and find that additional complexity need not improve performance. Klotz [14] relates numerical instability to formulation and finite precision. Optimal diagonal preconditioning is a broader problem: Qu et al. [17] use quasiconvex and semidefinite formulations, while Demmel [4] gives block-Jacobi guarantees. Our triangular calculation is an explanatory scalar special case, not a general preconditioning algorithm.

Robustness is also an established objective in MIP scaling. Berthold and Hendel [1] learn to select between standard and Curtis–Reid scaling in Xpress from an indicator of numerical difficulties. Their implementation uses power-of-two factors and their evaluation examines inconsistencies across seeds. Our external construction instead states row-level admissibility conditions; it does not predict or replace internal solver scaling.

Modeling environments already support generator-level information: Pyomo’s `ScaleModel` applies component factors [16], GAMS has variable and equation scale attributes [6], and IDAES derives and propagates scales across submodels [10, 11]. Plasmio.jl retains local problems and linking constraints [13]. Consequently, using modeling metadata before export is not new. Our identifiability result concerns the loss of one specific block statistic after a preceding normalization.

Tolerance-based scaling also has direct precedent. Gurobi documents absolute feasibility and integrality tolerances and the treatment of very small coefficients [7], and recommends external scaling that makes tolerance-sized violations negligible in application units [9]. Pritchett, Howell and Grebow [15] similarly scale constraints using physical accuracy requirements and SNOPT’s tolerance. Section 5 combines this known transfer with coefficient limits, identifies incompatible specifications, and solves the constrained centering problem. Integer rounding and exact binary64 export impose additional restrictions.

Finally, Cheung, Gleixner and Steffy [2] provide certificates for integer-programming results, and Hoen and Gleixner [8] analyze individual branch-and-bound decisions. Our rational residual checks concern primal acceptance; independent enumeration establishes optimality only on the small binary instances. The complementary attribution audit checks actual exports under matched kernels and coverage. These contributions concern identifiable mechanisms and explicit numerical contracts, without claiming universal performance superiority.

3 Model, policies and comparisons

Consider

$$\min c^T x \quad \text{subject to } Ax \leq b, \quad l \leq x \leq u, \quad x_j \in \mathbb{Z} \ (j \in J).$$

Table 1: Controls for the coefficient-only construction. Row eligibility and safeguards must also be matched for the indicated comparison. A role map and a block-scale map are separate information sources.

Policy	Local rows	Coupling rows	Information used
Base	None	None	None
Flat-GM	GM	GM	Row coefficients
Flat-L2	L2	L2	Row coefficients
Role-Hybrid	GM	L2	Row roles, coefficients
SA-Mat	GM	$\left[\sum_j (a_{rj}/s_{\beta(j)})^2\right]^{-1/2}$	Roles, block scales, ownership
SA-Pre	GM	Same expression, pre-local scales	Roles, original block scales

Equalities and opposite inequality senses are allowed. For a positive diagonal matrix D_r , replacing (A, b) by $(D_r A, D_r b)$ preserves every constraint, variable domain and objective value in exact arithmetic. It is essential to multiply the whole row, including its right-hand side, by the same positive number. Dividing different coefficients of a coupling row by different block scales is not a row scaling and is outside this equivalence statement.

Let the generator provide a fixed partition of the local rows into nonempty blocks, a coupling-row set, and a column-to-block map β . Coverage specifies which rows are eligible for scaling; a kernel maps eligible row data to a candidate factor; safeguards decide which factors are applied. These are separate choices. A block map can influence the factors even when coverage and kernels are fixed, but only if the block summaries retain information and are actually used.

For a nonzero row, put $m_r = \min_{a_{rj} \neq 0} |a_{rj}|$ and $M_r = \max_j |a_{rj}|$. The geometric-mean (GM) factor is $(m_r M_r)^{-1/2}$; the Euclidean (L2) factor is $1/\|a_r\|_2$. Both are coefficient-only rules. Calling them “the same kernel” because neither uses the RHS is incorrect. The following comparisons keep this distinction visible.

The direct test of the *block scales* is SA-Mat versus Role-Hybrid at matched coverage and safeguards. Flat-L2 also changes the local kernel, whereas Flat-GM changes the coupling kernel. Relabeling blocks without changing their assignments is an invariance check. Permuting the assignment of unequal scales is a different intervention: it tests the usefulness of the map on the chosen instances, and is not assumed to preserve semantic validity.

4 Identifying the contribution of structural metadata

An external scaling policy may use a row’s coverage status, a kernel selected by its role, and the ownership of its columns. These are distinct inputs. A comparison identifies the contribution of block ownership only when coverage, the row-role assignment, and the kernels remain fixed. Matching the local kernel alone does not meet this condition. This distinction has a particularly strong consequence for the post-normalization block statistic used by the original policy.

4.1 Collapse of post-normalization block centers

For each nonzero local row r , write m_r and M_r for its smallest and largest nonzero coefficient magnitudes. The coefficient geometric-mean kernel applies $d_r = (m_r M_r)^{-1/2}$. Let $\tilde{m}_r = d_r m_r$ and

88 $\widetilde{M}_r = d_r M_r$ denote the exported extrema. For a nonempty set I of local rows, define

$$s_I = \sqrt{\left(\min_{r \in I} \widetilde{m}_r\right) \left(\max_{r \in I} \widetilde{M}_r\right)}.$$

89 This definition uses all coefficients, including the row extrema.

90 *Proposition 4.1* (Exact collapse and the effect of a skip band). If every row receives its geometric-
 91 mean factor, then $s_I = 1$ for every nonempty grouping I of local rows. More generally, suppose
 92 the only safeguard is to replace d_r by 1 when its candidate belongs to $(1/u, u)$, where $u \geq 1$. Then
 93 $s_I \in [1/u, u]$, independently of the original coefficient scales and of the grouping of the rows.

94 *Proof.* Under exact normalization, $\widetilde{m}_r = \sqrt{m_r/M_r} = 1/\widetilde{M}_r$. Consequently $\min_r \widetilde{m}_r = 1/\max_r \widetilde{M}_r$,
 95 proving the first statement. For the second, let $c_r = \sqrt{\widetilde{m}_r \widetilde{M}_r}$. An accepted row has $c_r = 1$; a skipped
 96 row has $c_r = \sqrt{m_r M_r} \in (1/u, u)$. Thus $u^{-2} \leq \widetilde{m}_r \widetilde{M}_r \leq u^2$ for every row. Choose p attaining the
 97 maximum exported row magnitude and q attaining the minimum. Then

$$u^{-2} \leq \widetilde{m}_q \widetilde{M}_q \leq \widetilde{m}_q \widetilde{M}_p \leq \widetilde{m}_p \widetilde{M}_p \leq u^2.$$

98 Taking square roots gives the result. $\square$

99 For a coupling row a_r , let $\beta(j)$ assign column j to a local block, and define the block-weighted
 100 candidate and its ownership-blind counterpart by

$$\gamma_r^{\text{block}} = \left(\sum_j a_{rj}^2 / s_{\beta(j)}^2 \right)^{-1/2}, \quad \gamma_r^{\text{L2}} = \left(\sum_j a_{rj}^2 \right)^{-1/2}.$$

101 Blockless columns may use an arithmetic mean of valid block scales or the unit fallback; both
 102 conventions preserve the interval in the preceding proposition.

103 *Proposition 4.2* (Loss of ownership information). Under exact local normalization and with no
 104 coefficient filtering, $\gamma_r^{\text{block}} = \gamma_r^{\text{L2}}$ for every coupling row. Thus changing the block ownership map
 105 cannot change the candidate export. Applying the same coupling safeguards to the identical
 106 candidates preserves equality. Under the skip-band assumptions of Proposition 4.1,

$$u^{-1} \leq \frac{\gamma_r^{\text{block}}}{\gamma_r^{\text{L2}}} \leq u.$$

107 Before coupling safeguards, the condition numbers of the two exports differ by at most a factor u^2 ,
 108 on their common nonzero singular spectrum.

109 *Proof.* The equality follows by substituting $s_i = 1$. In the second case,

$$u^{-2} \|a_r\|_2^2 \leq \sum_j a_{rj}^2 / s_{\beta(j)}^2 \leq u^2 \|a_r\|_2^2,$$

110 which gives the factor bound. Write the block-based export as EA_{L2} , where E has unit entries on
 111 local rows and entries in $[1/u, u]$ on coupling rows. Multiplication by E changes each singular value
 112 by a factor between $1/u$ and u and preserves rank. Applying these inequalities to the largest and
 113 smallest positive singular values in both directions proves the condition-number bound. $\square$

114 Coefficient-window vetoes and filtering must be audited separately: they are outside the hy-
 115 potheses. Even exact row normalization does not preserve the reciprocity argument if the block
 116 summary discards extrema. For example, a row with magnitudes $(10^{-8}, 3, 10^8)$ is geometrically
 117 centered, but restricting the summary to $(10^{-6}, 10^6)$ produces $s_I = 3$. Such a nonunit statistic arises
 118 from the filter. It cannot, by itself, be interpreted as retained physical scale.

4.2 What a matched comparison identifies

With equal coverage and equal local kernels, two exports agree on their local rows. Their differences are confined to the coupling rows. This elementary localization result does not establish that the difference is due to block ownership: the coupling kernels may differ too. An appropriate ownership control keeps local geometric-mean normalization and coupling Euclidean normalization, but replaces all block scales by one. A separate role control compares this hybrid policy with Euclidean normalization on every row. Comparing only with geometric-mean normalization on every row mixes role-dependent kernel selection with ownership information.

Inactivity of one policy's coupling stage does not imply equality with a flat policy. Consider local rows $x_1 \leq 1$, $x_2 \leq 1$, and the coupling inequality $x_1 + 10^{-2}x_2 \leq 1$. Both local blocks have scale one. The Euclidean coupling candidate is $(1 + 10^{-4})^{-1/2}$ and is skipped by the band $(1/2, 2)$. The flat geometric-mean candidate is 10 and is applied. Both policies cover all rows, yet their coupling exports differ. Equality requires equality of the *actually applied* factors, rather than inactivity of just one policy.

4.3 A valid explanation of the stylized spectral slope

Take $B \geq 2$ local singleton rows with positive coefficients $v_i = 10^{e_i}$, where $\min_i e_i = -R$ and $\max_i e_i = R$, and one coupling row $v^\top$. Exact local normalization produces I_B and unit post-local block centers. A whole-row coupling factor γ gives

$$A_\gamma = \begin{bmatrix} I_B \\ \gamma v^\top \end{bmatrix}, \quad A_\gamma^\top A_\gamma = I_B + \gamma^2 v v^\top, \quad \kappa_2(A_\gamma) = \sqrt{1 + \gamma^2 \|v\|_2^2}.$$

Flat geometric-mean normalization selects $\gamma = 1$, whereas both the post-local block rule and the hybrid Euclidean control select $\gamma = 1/\|v\|_2$. Their condition numbers are therefore

$$\kappa_2(A_{\text{GM}}) = \sqrt{1 + \sum_i 10^{2e_i}}, \quad \kappa_2(A_{\text{block}}) = \kappa_2(A_{\text{hybrid}}) = \sqrt{2}.$$

For fixed B , the ratio is of order 10^{-R} . The slope is explained by the coupling kernel and survives removal of the ownership map. Every coupling coefficient is multiplied by the *same* γ , and all within-row ratios are preserved. Replacing v_i individually by $v_i/10^{e_i}$ would change the constraint and is not a row-scaling operation.

The matrix A_γ also illustrates a limitation of this spectral endpoint: letting γ tend to zero drives κ_2 to one while allowing an arbitrarily large original-row residual under a fixed exported residual budget. Thus minimizing this rectangular-matrix condition number alone does not enforce the intended numerical accuracy of the coupling constraint.

4.4 Exact scalar optimization in the triangular model

The idealized triangular model also admits a useful closed form that requires no ownership map after local whitening. It describes a common coupling factor, not the per-row optimum for a general generated MIP.

Proposition 4.3 (Optimal common coupling factor after local whitening). Let $R \in \mathbb{R}^{m \times n}$, with $m, n \geq 1$, and set $\rho = \|R\|_2$. For

$$C_\gamma = \begin{bmatrix} I_n & 0 \\ \gamma R & \gamma I_m \end{bmatrix}, \quad \gamma > 0,$$

the unique factor minimizing $\kappa_2(C_\gamma)$ is

$$\gamma_* = (1 + \rho^2)^{-1/2}, \quad \min_{\gamma > 0} \kappa_2(C_\gamma) = \sqrt{1 + \rho^2} + \rho.$$

With an admissible interval $0 < \ell \leq \gamma \leq h$, the optimum is $\min\{h, \max\{\ell, \gamma_*\}\}$.

The proof in Online Resource 1 reduces the spectrum by an SVD to two-by-two blocks. For the block at ρ , the singular-value ratio k satisfies $k + k^{-1} = \gamma^{-1} + \gamma(1 + \rho^2)$, whose unique minimizer gives the result.

For a single coupling row, γ_* normalizes that complete row's Euclidean norm, including the aggregate-variable coefficient. This provides an exact kernel explanation of the idealized conditioning improvement. It is not a guarantee about LP bases, presolve transformations, or mixed-integer runtime.

5 Equilibration with an explicit residual budget

Block ownership is not the only information a generator can supply. An application can also state how much violation of each original constraint is acceptable. Such a budget must come from the application or be declared as a numerical test parameter; it cannot be inferred reliably from a zero right-hand side or a block label. Scaling to make solver tolerances meaningful is established practice [7, 11]. The contribution of the following construction is an explicit feasibility test and an optimality characterization for combining that budget with coefficient limits, including a reason to abstain.

For a nonconstant row $a^T x \leq b$, define the original residual $v(x) = (a^T x - b)_+$ and specify a budget $\delta > 0$. Let $\epsilon > 0$ be a *hypothetical bound on the residual of the exported row*. A solver's feasibility parameter alone does not establish this hypothesis: internal scaling, presolve, integrality tolerances and finite precision intervene. For a positive row factor α , the identity

$$(\alpha a^T x - \alpha b)_+ = \alpha v(x) \tag{1}$$

holds in exact arithmetic. Consequently, the hypothesis of exported residual at most ϵ implies $v(x) \leq \delta$ whenever $\alpha \geq \epsilon/\delta$. Absolute residuals give the same identity for equality constraints.

Write $m = \min_{a_j \neq 0} |a_j|$ and $M = \max_j |a_j|$, and specify coefficient limits $0 < \ell \leq U$. These bounds concern the constraint matrix, not variable domains, objective coefficients or the right-hand side. If an RHS magnitude limit U_b is also required, intersect the upper bound below with $U_b/|b|$ for $b \neq 0$.

Proposition 5.1 (Feasibility and optimal constrained centering). *The factors that place every nonzero coefficient in $[\ell, U]$ and satisfy $\epsilon/\alpha \leq \delta$ form exactly the interval*

$$\mathcal{I} = [L, H], \quad L = \max\{\epsilon/\delta, \ell/m\}, \quad H = U/M. \tag{2}$$

The interval is nonempty if and only if

$$M/m \leq U/\ell \quad \text{and} \quad \delta \geq \epsilon M/U. \tag{3}$$

When it is nonempty, the unique minimizer of

$$\min_{\alpha \in \mathcal{I}} \max_{a_j \neq 0} |\log(\alpha |a_j|)| \tag{4}$$

is

$$\alpha^* = \min\{H, \max\{L, (mM)^{-1/2}\}\}. \tag{5}$$

If $L > H$, no positive row factor meets both requirements.

Proof. The coefficient bounds are equivalent to $\alpha m \geq \ell$ and $\alpha M \leq U$. Together with $\alpha \delta \geq \epsilon$ they give (2). The condition $L \leq H$ is equivalent to the two inequalities in (3). To solve the objective, put $t = \log \alpha$, $q = (\log m + \log M)/2$ and $h = (\log M - \log m)/2$. Since the extreme absolute logarithms occur at an endpoint,

$$\max_j |t + \log |a_j|| = \max\{|t + \log m|, |t + \log M|\} = h + |t + q|.$$

The unique minimizer on $[\log L, \log H]$ is the projection of $-q$ onto that interval. Exponentiating gives (5). $\square$

Thus the rule optimizes a coefficient-centering criterion; it does not minimize a condition number or predict running time. The two causes of abstention have different interpretations. A within-row ratio exceeding U/ℓ cannot be repaired by row scaling. A budget below $\epsilon M/U$ conflicts with the prescribed coefficient ceiling even if the row is perfectly balanced. Returning the base row does not resolve either conflict and must not be reported as satisfying the contract. The test identifies an incompatibility among numerical specifications, not an infeasible optimization model. Revising the units, limits or formulation requires a separate decision.

Corollary 5.2 (Invariance under positive re-emission). *Replace (a, b, δ) by $(qa, qb, q\delta)$ for any $q > 0$, keeping ϵ, ℓ, U fixed. The interval becomes $\mathcal{I}/q$, abstention is unchanged, and the selected factor becomes α^*/q . The exported row and the budget-normalized original residual $v(x)/\delta$ are therefore unchanged.*

Proof. Both coefficient extremes and the original budget acquire the factor q . Each term defining L , H and $(mM)^{-1/2}$ is divided by q . Equation (1) gives the residual statement. $\square$

This invariance is lost if the row is re-emitted but its budget is not transformed with it. It also distinguishes a budget in canonical units from a fixed cutoff on the scaling factor itself. Restricting factors to powers of two can avoid additional rounding in multiplication when no underflow or overflow occurs, but this familiar technique [1] preserves the stated invariance only for power-of-two re-emissions, subject to representability. The dyadic construction below explicitly checks those arithmetic restrictions.

5.1 The part that row scaling cannot fix: integer rounding

Let J be the integer columns and let $\bar{x}$ be obtained by rounding only these coordinates of x . Suppose $|\bar{x}_j - x_j| \leq \eta_j$ for $j \in J$ and put $E = \sum_{j \in J} |a_j| \eta_j$. The triangle inequality and (1) give the useful but conditional bound

$$v(\bar{x}) \leq \epsilon/\alpha + E. \tag{6}$$

It applies to equalities with absolute residuals as well. The two terms have different behavior: changing the row factor can reduce the exported-residual term, whereas it leaves the amplification of integer rounding in original units unchanged. The scalar normalization of a big- M row therefore cannot, by itself, remove the effect of an integrality tolerance on that row.

For a contract based on the worst-case bound (6), Proposition 5.1 applies after replacing δ by $\delta - E$, provided $\delta > E$. If $\delta \leq E$ and $\epsilon > 0$, no positive factor satisfies this worst-case contract. This does not say that every returned solution will violate its budget: actual rounding errors may be smaller or cancel, and other constraints can exclude the worst case. It says that the stated information is insufficient to certify it. Re-emission invariance still holds because E transforms to qE . A tighter integrality bound or a reformulation can reduce E ; pure row scaling cannot. This distinction prevents the row-only guarantee from being mistaken for a guarantee about a rounded MIP solution.

5.2 An exact export on a dyadic grid

Even when a factor satisfies the real-arithmetic interval, independently rounded products $\text{fl}(\alpha a_j)$ and $\text{fl}(\alpha b)$ need not describe an exactly proportional row over the stored data. Merely rounding a computed endpoint to nearest can also move it outside a closed admissible interval. Our arbitrary-factor implementation forms the interval endpoints as exact rational numbers in the supplied binary64 inputs and rounds them *inward*; it abstains if the interval contains no positive binary64 factor. This certifies factor admissibility, not exactness of all subsequent products.

Powers of two provide an established way to avoid multiplication rounding. The following explicit intersection also accounts for subnormal numbers and for the RHS. Assume finite binary64 input data and IEEE arithmetic with gradual underflow [12]. Write each nonzero datum $z \in \{a_1, \dots, a_n, b\}$ uniquely as $z = \text{sign}(z)N_z2^{p_z}$ with N_z a positive odd integer, and put $q_z = \lfloor \log_2 |z| \rfloor$. Let $[L, H]$ be the desired real factor interval, including any rounding allowance from (6).

Proposition 5.3 (Exact dyadic admissibility). *A factor $\alpha = 2^k$ is positive and representable in binary64, belongs to $[L, H]$, and multiplies every nonzero coefficient and the RHS exactly into a finite binary64 value if and only if the integer k satisfies*

$$K_- = \max \left\{ -1074, \lceil \log_2 L \rceil, \max_z (-1074 - p_z) \right\} \leq k, \\ k \leq K_+ = \min \left\{ 1023, \lfloor \log_2 H \rfloor, \min_z (1023 - q_z) \right\}.$$

If $K_- > K_+$ there is no such factor. Otherwise an exponent nearest to $-\frac{1}{2} \log_2(mM)$ among the integers in $[K_-, K_+]$ minimizes the logarithmic coefficient objective (4) on this exact-export set. A tie may be resolved by choosing the larger factor.

Proof. The factor itself requires $-1074 \leq k \leq 1023$. Since the datum's odd significand is unchanged, the product $N_z2^{p_z+k}$ lies on the subnormal grid exactly when $p_z + k \geq -1074$; its highest bit stays below overflow exactly when $q_z + k \leq 1023$. The original significand has at most 53 bits, so these two conditions are also sufficient. Intersecting them over the row data and with $L \leq 2^k \leq H$ gives the interval of integer exponents. The objective is $\frac{1}{2} \log(M/m) + |k \log 2 + \frac{1}{2} \log(mM)|$, proving the nearest-exponent claim. $\square$

The implementation obtains the endpoint exponents using integer ratios and bit lengths, and compares candidate centering objectives exactly as rational numbers. It also checks the rounded products against exact rational products. A round-trip decimal serializer is necessary if the row is subsequently written to a text file: fixed decimal places can discard small values even when the multiplication is exact. These checks concern the represented mathematical model. They still do not guarantee that the solver will return an acceptable point.

Verification after solving. The interval theorem is conditional and exact-arithmetic. Every returned point must still be checked against the original rows, bounds and integrality restrictions. Our small stress experiments compute row residuals by exact rational arithmetic on the supplied binary64 coefficients and returned solution. They also enumerate the binary assignments to determine the optimum independently. These checks are possible because those instances are deliberately small. On large dispatch models the archived checker uses floating-point residuals, which are diagnostics rather than a proof of MIP optimality. General certificates of branch-and-bound correctness address a much stronger task [2].

6 Computational design

6.1 Matrix attribution audit

The C++ generator constructs local blocks and coupling rows while retaining their row roles and column ownership. The archived headline cell uses four heterogeneous blocks, coefficient exponent range $[-2, 2]$, coupling disturbance severity 6, and 30 seeds. All policies in [table 1](#) are available in the same implementation. Their exported matrices allow a comparison before any optimizer is called. The analysis records spectral pseudo-condition numbers and direct export comparisons; equal condition numbers alone are not evidence of identical matrices.

The spectral diagnostic is $\kappa_2^+(A) = \sigma_{\max}(A)/\sigma_{\min}^+(A)$, where numerical rank is determined relative to the largest singular value. It is a diagnostic of the exported constraint matrix, not of the active basis at every node. Variable bounds, redundant rows and slack conventions can change its interpretation. A condition-number ratio is never used as a proxy for verified solver success.

6.2 Operational reconstruction

The operational case study uses the final four-arm campaign from the dispatch repository: Base, Flat-GM, Flat-L2 and Role-Hybrid; 15 instances from each of the Simple and SG-Ter-Mer families; Gurobi; relative MIP gaps 1% and 0.1%; seed 42; 16 threads; and a 1,800-second solver limit. The fixed subsets are retained rather than reselected after inspecting outcomes. This gives 240 assigned runs. We retrieved all native logs from the cluster and preserved input hashes.

The primary runtime endpoint is arithmetic mean PAR10 over all 15 assigned instances per cell, with 18,000 seconds assigned to a non-completion. This measures solver completion rather than independently certified feasibility. Native logs recover Work at a common resolution of 0.01 units; the archived runner had accidentally rounded scaled-variant markers to integers. Work is analyzed separately on the common all-variants-completed pool, with exploratory paired bootstrap intervals. Online Resource 1 specifies the reconstruction, resolution sensitivity and residual checks. The selected pools and repeated instances across gap settings do not constitute independent replications of general solver performance.

6.3 Enumerated residual-budget stress experiment

We construct three binary families with 12 variables: a cardinality row, three positive packing rows, and three rows with signed integer coefficients. Profits are positive integers. If W is the integer matrix and k an integer capacity vector, the canonical constraints are

$$10^e Wx \leq 10^e k - 0.1 \mathbf{1}, \quad x \in \{0, 1\}^{12}, \quad e \in \{0, 3, 6, 9\}. \quad (7)$$

For these exponents the exact binary feasible set is $Wx \leq k - \mathbf{1}$. Thus changing e changes the numerical representation and continuous relaxation while preserving the binary feasible set. It is not pure row re-emission, because the absolute margin 0.1 remains fixed. The corresponding distinction is intentional: it exposes integer acceptance near a facet without changing the binary optimum.

The budget is 0.01 in the canonical row's units. These are declared stress-test units, not engineering tolerances inferred from the dispatch data. Eight seeds, 100–107, were reserved after a one-seed exploratory pilot; all four exponents and all three families are retained. Both presolve settings are run on HiGHS 1.15.1 and Gurobi 13.0.2, with one thread, zero target MIP gap, a ten-second limit, and primal/integrality tolerances 10^{-6} . The nine policies give 3,456 planned configurations. Budget-round abstains on 288 of them, leaving 3,168 optimizer calls.

Base, GM and L2 have their definitions in table 1, with full coverage and no identity-band veto in this experiment. GM-tight uses GM and tightens both tolerance settings to 10^{-9} . Budget-only uses the established direct factor ϵ/δ_r . Budget-GM uses proposition 5.1, coefficient limits $[10^{-9}, 10^9]$ and $\epsilon = 10^{-6}$. Budget-round additionally reserves $E_r = 10^{-6} \sum_j |a_{rj}|$ for rounding and abstains on the whole model if any row has an empty interval. Abstention is counted separately, not as a successful solve. Dyadic-GM uses a power-of-two GM factor with coefficient limits and an exact binary64 export; Budget-dyadic additionally imposes the row budget through Proposition 5.3. These two arms were added after inspecting the floating-point endpoint issue in the initial run. They are an explicitly exploratory arithmetic sensitivity, not an independent holdout. The original seven arms are retained with the corrected endpoint implementation and unchanged instance seeds.

All 2^{12} assignments are enumerated to obtain the binary optimum. The script checks that each stored binary64 RHS lies strictly between the two adjacent scaled integer activities used in that argument. Returned solutions, row factors and generated data are saved. Original row residuals are evaluated as exact rational expressions in the stored binary64 data. A verified optimum requires a feasible rounded binary assignment with the enumerated objective, valid original variable bounds, and a returned point within 10^{-9} of its rounded assignment. We also record residual budgets and integrality errors separately. This conservative acceptance rule is declared independently of the solver’s status; a rejected approximately integral point is not automatically a solver implementation error.

This experiment tests a failure mechanism. It is too small and deliberately too close to integer feasibility boundaries to estimate practical runtime or failure rates on industrial MILPs. Counts are descriptive, and no significance tests treat the repeated solver settings on a generated instance as independent samples.

7 Results

7.1 A kernel effect mistaken for a block effect

In the archived heterogeneous headline cell, the median spectral diagnostic is 5.30×10^8 for Base, 1.36×10^4 for Flat-GM, 1.33×10^3 for SA-Mat and 4.68×10^2 for Flat-L2. The ratio of Flat-GM to SA-Mat is about 10.2. That comparison changes the coupling kernel. Role-Hybrid matches SA-Mat in the 30 archived spectral records, and permuting the block-scale assignments leaves the same diagnostic. The collapse result explains why the block-relative formula becomes an ordinary L2 normalization in this regime. Flat-L2’s smaller diagnostic does not establish a better MIP policy: it changes local normalization as well and requires a separate solve/verification evaluation.

Using pre-local block scales does not rescue a positive metadata claim on these instances. SA-Pre is worse than Role-Hybrid on all 30 archived instances, with a median paired condition-number ratio about 4.27. In the exhaustive 24-permutation assignment sweep, the true scale assignment is never the best; its median rank is 18. These are falsification results for this choice of summary and benchmark. They are not a theorem that semantic metadata or all block-based preconditioners are useless.

Fresh exports with matched controls. The new replication uses ten seeds, 20260909–20260918, at each of three exponent ranges, two severities and three coupling patterns: 180 generated instances and 1,080 exports. The corrected serializer preserves binary64 values on readback, including coefficients, RHS values and bounds. All exports preserve named structure and pass the whole-row proportionality check at relative tolerance 10^{-13} . This is a floating-point transformation audit, not an exact rational equivalence certificate.

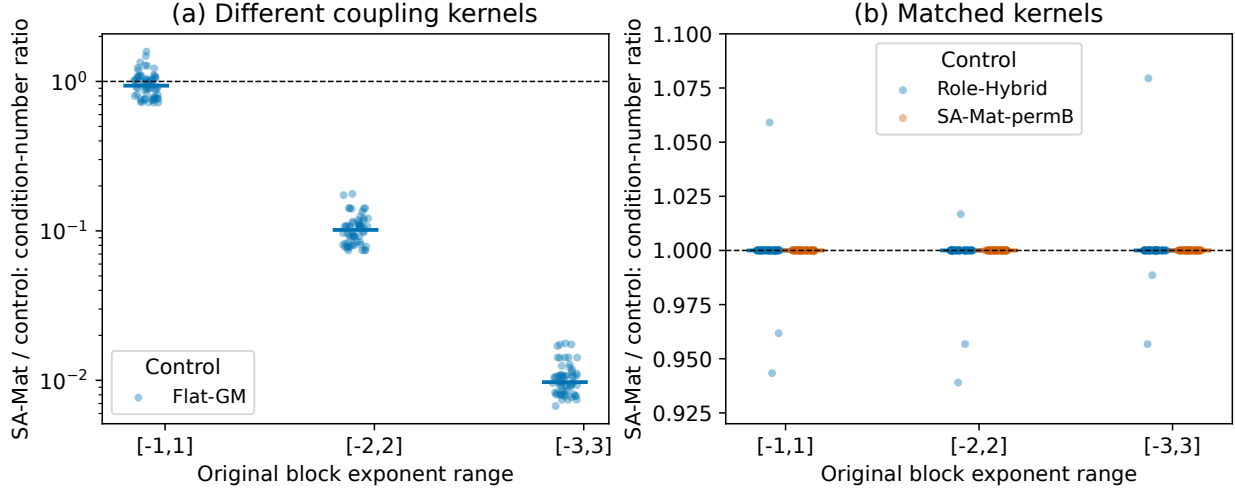


Figure 1: Fresh matrix audit. Each exponent range contains 60 generated instances. The left comparison changes the coupling kernel; the right comparisons hold the kernels fixed and remove or permute block-scale information. Marks show individual ratios with deterministic horizontal jitter; short horizontal segments are medians. The panels use different vertical scales to expose the remaining small differences.

SA-Mat and Role-Hybrid are byte-identical on 136 of 180 instances; SA-Mat and its permuted-scale counterpart are byte-identical on 165. The nonidentical cases matter: the production implementation includes window vetoes and filtered block extrema, so the exact-collapse hypotheses need not hold universally. On the common-rank spectral diagnostic, the SA-Mat/Role-Hybrid ratio lies between 0.939 and 1.079, with median one. The scale-permutation ratio lies between 0.99962 and 1.00022. In contrast, the median SA-Mat/Flat-GM ratios are 0.937, 0.102 and 0.00973 as the exponent range grows from $[-1, 1]$ to $[-3, 3]$. Thus the large heterogeneity-dependent advantage is explained by the coupling kernel; the matched control reveals smaller implementation-dependent differences.

Numerical rank is fixed per instance from the row-L2-normalized Base at threshold 10^{-12} and used for every variant. Independently thresholding each variant would give different ranks in 25 of the 1,080 exports. Several unscaled matrices approach binary64 resolution, so their very large condition estimates are not interpreted quantitatively. The scaled matrices used in the paired ratios have diagnostics below 6.5×10^6 . Direct named-export comparisons carry the identity claims and do not depend on singular-value estimation.

7.2 Completed-run gains and timeout-sensitive outcomes

On the 14 common completed SG-Ter-Mer instances, Flat-GM/Base has Work geometric-mean ratios 0.698 and 0.682 at the two gaps. Yet instance077 completes under Base and times out under every scaled policy at both gaps, sharply reversing the fixed-pool PAR10 ordering (Online Resource 1, Table S1). The Simple family also has no uniform ordering: Flat-GM reduces the arithmetic runtime summary at 1% but increases it at 0.1%. These results illustrate why completed-run effort and failure-sensitive runtime must be reported separately.

Original-data residuals also require an explicit interpretation. An absolute equality residual of 4.97 on a zero-RHS row in caso46e is not a 497% physical error; the same run's maximum activity-

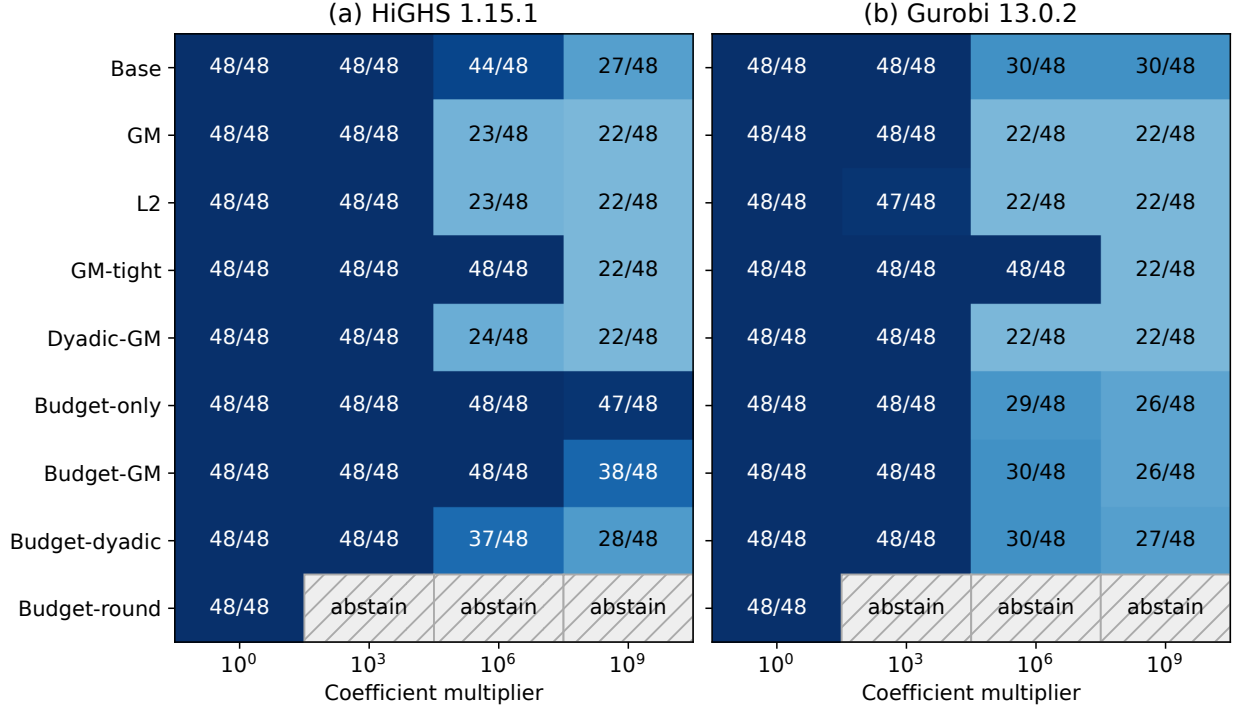


Figure 2: Verified rounded optima divided by planned configurations, by coefficient multiplier. Each cell contains 48 configurations (three families, eight seeds and two presolve settings) per policy and solver. The fixed margin in (7) makes the continuous feasibility boundary increasingly close to an integer activity. Hatched cells are explicit abstentions.

normalized residual is 1.15×10^{-7} . Online Resource 1 gives both definitions and the exploratory Work intervals. Neither residual normalization establishes application acceptance without a justified original-unit budget.

7.3 Conditional contracts and verified binary optima

The full counts appear in Online Resource 1 (Table S2). Budget-GM verifies 182 of 192 configurations on HiGHS, compared with 141 for GM and 167 for Base. Budget-only verifies 191: direct budget scaling is more successful than the extra minimax centering in this experiment. On Gurobi, Budget-GM verifies 152, compared with 140 for GM and 156 for Base. GM-tight verifies 166 under each solver. Thus the row-budget construction does not establish a solver-independent reliability improvement, and its conditional hypothesis must remain visible.

Dyadic-GM verifies 142 configurations on HiGHS and 140 on Gurobi; Budget-dyadic verifies 161 and 153, respectively. Exact representation of the scaled row is therefore insufficient for reliable solver acceptance on these stress cases. In a preserved earlier implementation, rounding the budget endpoint to nearest instead of inward changed HiGHS Budget-GM’s verified count from 182 to 191. This is a numerical sensitivity result, not a reason to prefer a factor that fails its declared interval. The artifact preserves both implementations’ outputs and identifies the corrected run.

Budget-round accepts and verifies 48 configurations per solver, and abstains on the other 144. Its zero observed verification failures have a substantial cost: it declines many instances that another policy solves correctly. This is expected of a worst-case rounding bound. An incompatibility

certificate explains why the requested uniform contract cannot be asserted; it is not evidence that the MILP itself cannot be solved. Figure 2 reports the denominator explicitly so that abstention cannot masquerade as perfect overall performance.

8 Discussion and limits

The collapse theorem concerns extrema-based summaries after geometric-mean normalization. It does not rule out useful variable scales, physical tolerances or other block statistics. Its practical implication is to hold coverage, kernels and safeguards fixed before attributing an effect to block information, and to inspect the actual row factors or named exports. Safeguards can introduce discontinuities; equal scalar diagnostics or rounded Work values cannot prove export identity.

Residual budgets give a distinct interpretation to generator information. Their joint compatibility with coefficient limits is explicit and invariant under consistent row re-emission. The contract remains conditional: integer-rounding amplification and internal solver transformations can defeat predictions based on the exported row alone. Production use requires a postsolve acceptance or repair policy and measurement of its full cost.

The dispatch reconstruction covers two application classes, one solver, one seed and 15 selected instances per class. The new budget policies have not been validated there using application-supplied engineering tolerances. The matrix audit tests an algebraic mechanism; the binary instances deliberately amplify tolerance sensitivity. Neither estimates typical industrial failure rates or speedups. Enumeration certifies the small-instance optima, with no corresponding exact optimality claim for dispatch.

9 Conclusion

For the studied local-block-coupling construction, geometric-mean row centering can remove all information from a subsequent extrema-based block-scale summary. A bounded identity band only restores bounded scale variation. This explains why changing a coupling-row kernel can look like a block-metadata benefit when the comparison does not hold kernels fixed. The operational reconstruction further shows that favorable completed-run effort can coexist with an unfavorable failure-sensitive runtime summary.

An explicit residual budget provides a separate, interpretable basis for row scaling. Its compatibility with coefficient limits can be characterized exactly, and constrained logarithmic centering has a closed-form solution. Integer-rounding amplification and solver-internal transformations limit what this conditional contract can guarantee. The resulting recommendation is to expose the information used, audit the actual transformation, declare incompatible specifications, and verify the returned point.

Data and software availability

The accompanying repository contains the C++ generator, frozen historical aggregates, the reconstructed operational logs and analysis, the exact-enumeration experiment, row-factor implementation, tests, and scripts generating the new tables and vector figures. The current research sources are distinguished from the historical draft, whose positive block-attribution claim is superseded. All newly reported stress data and solutions are stored under `results-revision/research-audit/`. No commercial solver is required to regenerate figures, verify the saved binary solutions, or run the HiGHS stress arm. Re-running the Gurobi arm requires a suitable license.

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Authorship and computational resources. Mathias Rodríguez Castro is the sole author. The computational platform originates in his hydrothermal-dispatch thesis work; the operational experiments used ClusterUY.

Use of AI assistance. OpenAI Codex was used to assist with mathematical derivations and their internal review, experiment and verification code, literature analysis, and drafting and revising the manuscript. Reported numerical results come from the stored computational runs. Figures were generated programmatically from those data or the stated equations. The author is responsible for the scientific content and the version submitted for publication.

Supplementary information

Online Resource 1 provides the triangular-model proof, operational reconstruction details, frozen experiment protocols, arithmetic implementation checks, and reproduction instructions. The accompanying source and data archive is Online Resource 2.

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